

LEVEL 1 REPORT

Task 1: Data Cleaning and Exploratory Data Analysis (EDA)

1. Introduction

This task focused on performing data cleaning and exploratory data analysis on the Iris dataset using R. The objective was to identify data quality issues, compute summary statistics, visualize distributions, and analyze relationships between numerical variables.

2. Data Cleaning and Preprocessing

- The dataset was successfully loaded using `read.csv()`.
- The structure of the dataset was examined using `str()` and `head()`.
- Missing values were checked using `colSums(is.na(rr))`.
 - Result: No missing values were found in the dataset.
- Duplicate rows were removed using `duplicated()`.
 - Result: No duplicate records were present.
- The species column was converted to a categorical variable (factor) for proper analysis.

The dataset was clean and required minimal preprocessing.

3. Summary Statistics

The summary statistics revealed:

- Sepal Length
 - Mean: 5.856
 - Standard Deviation: 0.829
- Sepal Width
 - Mean: 3.056
 - Standard Deviation: 0.437
- Petal Length

- Mean: 3.78
 - Standard Deviation: 1.759
- Petal Width
 - Mean: 1.209
 - Standard Deviation: 0.758

Petal length showed the highest variation among the features.

4. Data Visualization

The following visualizations were created:

- Histogram of Sepal Length (distribution analysis)
- Boxplot of Sepal Length (outlier detection)
- Scatter Plot of Sepal Length vs Petal Length (relationship analysis)

The scatter plot showed a strong positive relationship between sepal length and petal length.

5. Correlation Analysis

The correlation matrix showed:

- Strong positive correlation between:
 - Petal Length and Petal Width (0.96)
 - Sepal Length and Petal Length (0.87)
- Weak negative correlation between:
 - Sepal Width and Petal Length (-0.42)

This indicates that petal features are strongly related and may be important for classification tasks.

6. Conclusion

The dataset was clean with no missing values or duplicates. Exploratory analysis revealed strong correlations among petal measurements, suggesting they are key distinguishing features in the dataset.

VISUALITATIONS



